CLINICAL TRIAL PATIENT MATCHING

DOMAIN: HT-12

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PROBLEM SCENARIO

Clinical trials struggle to recruit eligible patients efficiently. Researchers rely on manual database searches and outreach Patients are often unaware of nearby relevant trials. Trial eligibility involves complex inclusion/exclusion criteria. Leads to delays in drug development and increased costs

CURRENT PAIN POINTS

- ⏳ Slow, manual patient screening process
- ❌ High risk of human error and bias in matching
- 🔍 Difficulty handling complex criteria (age, genetics, history)
- 🔒 Privacy concerns when sharing patient data
- 📉 Low patient awareness and engagement
- 🗂️ Data silos across hospitals (no unified system)
- 📊 Incomplete patient records affecting decisions

OBJECTIVES &SUCCESS

- Objective:

Build a Clinical Trial Matching Engine to automate patient-trial matching with privacy protection

- Success Criteria:

✅ Accurate matching based on eligibility rules 🔐 Strong data anonymization before consent ⚡ Faster recruitment time 📈 Increased trial participation rate 👤 Improved patient awareness and engagement

SOLUTION STRATEGY

Develop a centralized matching platform. Use rule-based + intelligent filtering engine. Allow researchers to define criteria in structured format (JSON)Implement Privacy Shield (anonymized patient data)Notify patients about relevant trials automatically Enable multi-hospital integration. Support partial matching for incomplete data

SYSTEM ARCHITECTURE

Components:

- 👩‍⚕️ Patient Portal (profile + medical history)
- 🔬 Researcher Portal (trial creation + criteria)
- ⚙️ Matching Engine (core logic)
- 🔐 Privacy Layer (data anonymization)
- 📩 Notification Service
- 🗄️ Database (patients, trials, logs)
- 🌐 Integration Layer (external registries)Architecture

Style:

Microservices-based (scalable & modular)API-driven communication

DATA FLOW/ WORKFLOW

Patient registers and uploads medical data.
Researcher creates trial with eligibility criteria.
Matching engine processes patient data vs criteria. Eligible patients identified (fully or partially). Data anonymized before sharing with researchers. Patients receive notifications.
Patient gives consent → full data shared
Recruitment tracked in dashboard

IMPLEMENTATION AND DEMO REFERENCE

Tech Stack :

Frontend: React / Angular

Backend: Node.js / Django

Database: MongoDB / PostgreSQL Matching

Logic: Rule-based filtering (hard exclusions)Weighted scoring (flexible criteria)APIs for integration with trial registries

Demo: Show patient signup → matching → notification → consent flow

IMPACTS AND BENEFITS

- 🚀 Faster clinical trial recruitment
- 🎯 Improved matching accuracy
- 🔐 Enhanced patient privacy
- 📊 Better decision-making for researchers
- 🌍 Increased accessibility to trials
- 💊 Accelerates drug development lifecycle

LIMITATIONS

- ⚠️ Dependence on data quality
- ⚠️ Integration challenges across hospitals
- ⚠️ Complex rule definitions may need tuning
- ⚠️ Regulatory compliance (HIPAA/GDPR-like concerns)
- ⚠️ Initial adoption resistance

FUTURE SCOPE AND ROADMAP

- 🤖 AI/ML-based predictive matching
- 🧬 Integration with genetic data systems
- 🌍 Global hospital network expansion
- 📱 Mobile app for patient engagement
- 🗣️ Multilingual support
- 📈 Advanced analytics dashboard
- 🔗 Real-time sync with global trial registries